

1. Show the topology of the Type-I and Type-II transmembrane proteins (how they are oriented toward cytoplasmic and exoplasmic side of the membranes). (3)
2. Discuss the mechanism of co-translational translocation of secretory proteins across the endoplasmic reticulum membrane. (10)
3. Schematically discuss one of the mechanisms by which a protein is transported to the inner mitochondrial membrane. Consider this protein to be a single-pass membrane protein of inner mitochondrial membrane. (7)
4. What is the role of the SNARE proteins in the vesicular traffic? (4)
5. What is the role of Dynamin in the trafficking of clathrin-coated vesicles? (2)
6. Let us assume that a ligand L induces a specific signalling pathway in the target cell. This specific signalling pathway involves interaction of a signalling protein A with another signalling protein B. Design an experiment using a co-immunoprecipitation-based assay to determine that indeed protein A interacts with protein B, upon exposure of the cells to the ligand L. (6)
7. Treatment of mice with a specific reagent X increases T-helper (Th) cell population. Let us assume that Thy1.2 and CD4 are the two specific markers of Th cells. How will you experimentally determine this using a flow cytometry-based assay. Show the representative plots of your flow cytometry assay. Principles of flow cytometry are not required to be described. (6)
8. The outer mitochondrial membrane (OM) and inner mitochondrial membrane (IM): A. does not contain lipid. B. is more permeable as compared to inner membrane due to presence of porin. C. sets up a proton gradient. D. swells and contracts readily. Select and write the correct feature of the OM. Select and write the correct statement for IM. (2)
9. An mRNA coding for a secretory protein, when translated using the free ribosome under in vitro conditions, resulted in a 40 kDa protein. The same mRNA when translated using the rough endoplasmic reticulum, resulted in a 36 kDa protein. The difference in the molecular weight of the two polypeptides is due to the loss of a: A. 2 kDa peptide from the N-terminus and 2 kDa peptide from C-terminus. B. 1 kDa peptide from N-terminus and 3 kDa peptide from C-terminus. C. 4 kDa peptide from N-terminus. D. 4 kDa peptide from the C-terminus. Select and write the correct option. (2)
10. Proteins are transported in the cell through the membranous vesicles. These vesicles: A. form when a section of membrane protrudes and buds off. B. have a layer of coat proteins around the inside of the vesicle. C. Use their transmembrane v-SNARE protein to find their target membrane. D. move with the help of motor proteins. Select and write all the correct statements. (2)
11. Golgi-associate vesicles transport: A. ER resident proteins back to ER. B. Golgi resident proteins in both retrograde and anterograde directions. C. Cargo proteins in an anterograde direction. D. All of the above statements are correct. Select and write the correct option. (1)
12. Which of the following vesicles does not have a clathrin coat when it is first formed? A. transport vesicles that form from the ER. B. vesicles formed from the coated pits on the plasma membrane. C. Transport vesicles that take cargo from the trans-golgi. D. Vesicles that take cargo to the sorting vesicles (late endosome). (1)
13. Which of the following statement(s) is/are false about transport vesicles? A. COPII vesicles transport proteins from rough ER to golgi. B. Clathrin-coated vesicles transport proteins from the trans golgi to late endosomes. C. COPII vesicles transport proteins between Golgi cisternae and from cis-golgi back to ER. (1)
14. In eukaryotes, which one of the following is not a second messenger? A. 1,2-diacyl glycerol. B. cyclic AMP. C. inositol 1,4,5-triphosphate. D. calcium ions. E. triacylglycerol. (1)
15. Which of the following biochemical reactions is most commonly used by living cells to propagate intracellular signals? A. Acylation. B. Methylation. C. Glycosylation. D. Phosphorylation. E. Decarboxylation. (1)
16. Diacylglycerol activates: A. protein kinase A. B. protein kinase C. C. MAP kinase. D. tyrosine kinase. (1)